FastSpeech 2-Based English and Telugu Text-to-Speech

1. Introduction

This project explores FastSpeech 2-based text-to-speech synthesis for English and Telugu, including dataset preparation, speech generation and variance-adaptor analysis.

2. FastSpeech 2 Architecture

FastSpeech 2 uses a text encoder, variance adaptor and mel-spectrogram decoder. The variance adaptor handles duration, pitch and energy information. A neural vocoder converts the generated mel spectrogram into speech audio.

3. English Dataset and Implementation

The English dataset is Svarah. The inspected preprocessed dataset contains 2,815 training entries and 557 validation entries, with matching mel, pitch, energy and duration feature files. Five English test audio files were retained. The dataset is available from the Kaggle source linked in the repository.

Status: A verified trained English FastSpeech 2 acoustic checkpoint and training/validation loss curves are not available in the current backup. The English audio is not claimed as output from a verified trained checkpoint.

4. Telugu Dataset and Implementation

The Telugu dataset source is IndicTTS Telugu. The prepared subset contains 2,036 training utterances (approximately 5 hours) and 376 validation utterances (approximately 1 hour). Audio was resampled to mono 22,050 Hz WAV, and matching WAV/LAB pairs were verified.

Five Telugu test samples were generated using the pretrained Telugu female FastSpeech 2 model from the Fastspeech2_HS project. A HiFi-GAN vocoder was used for waveform generation. This is third-party pretrained inference, not training from scratch for this assignment.

5. Test Samples and Variance Adaptor Analysis

Telugu samples T1, T2a, T2b, T3 and T4 were generated. For each sample, duration, pitch and energy predictions were extracted from the model inference output. The values were saved as NumPy arrays and visualized in plots. These are token-level model predictions, not ground-truth alignment values. For T4, date and time numerals were expanded into Telugu words for synthesis.

6. Results

Five English audio files and five Telugu audio files are retained in the project results folders. Five Telugu variance-adaptor figures and five prediction-data files are also retained.

7. Limitations

The available artifacts do not establish successful from-scratch training of both language models. The English acoustic checkpoint and verified loss curves are unavailable. The Telugu model is pretrained. Therefore, these results are documented as dataset preparation and inference artifacts rather than claiming that both assignment models were independently trained.

8. References

English Svarah: Kaggle dataset
https://www.kaggle.com/datasets/sirichandana6281/fastspeech2-english-training
Telugu IndicTTS: https://huggingface.co/datasets/SPRINGLab/IndicTTS_Telugu
Telugu pretrained model: https://github.com/utkarsh2299/Fastspeech2_HS